

Friterie Riebedebie Report

Operation analytics

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PART A : Queuing management

1. Discuss the customer-introduced variability at the Fries shop. Identify the most likely sources of variability.

In a fast-food setting like Friterie Riebebedie, customer-introduced variability plays a significant role in the efficiency of operations. The most common sources of variability include arrival times, order sizes, service times, and customer behavior.

- **Customer arrival variability is a major factor**. Customers do not arrive at a constant rate throughout the day. There are peak hours (such as lunchtime and evening) when demand surges, and off-peak hours with very few customers. This creates a mismatch in workload, making it difficult to maintain an optimal staffing schedule.
- **Order size variability affects service efficiency**. Some customers may order just a portion of fries, while others may request large meals with multiple items. Larger orders take more time to prepare, leading to longer service times and increased queue lengths during busy periods.
- **Service time variability is another crucial factor**. Not all orders take the same time to prepare. Even with standardized preparation methods, different types of food require varying cooking times. Additionally, new or slower employees may take longer to fulfill orders compared to experienced staff.
- **Customer behavior variability** can **impact queue dynamics**. Some customers may ask additional questions, request modifications to their orders, or take longer to pay. Others may decide to abandon the queue if the wait time is too long, leading to lost sales. These unpredictable actions make it difficult to predict and control service flow efficiently.

To manage this variability, **effective staffing adjustments, queue management strategies, and service process optimizations are essentia**l. The goal is to reduce waiting times and lost customers while maintaining high service efficiency.

2. What is the capacity utilization at Friterie Riebebedie, and how is this affecting waiting times and the proportion of lost customers? Perform a queuing analysis.

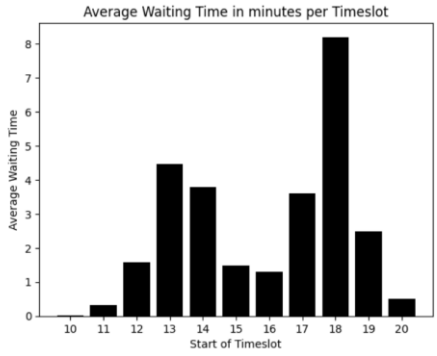
Waiting time and % rejected customers per time slot

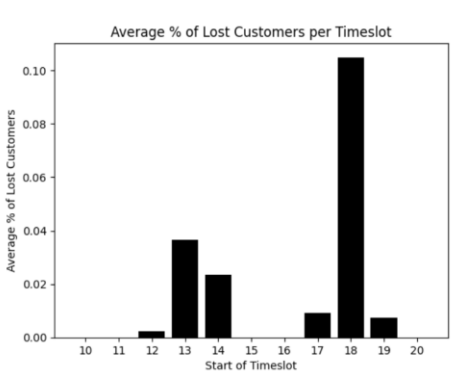
- From the original analysis, we observed that waiting times were significantly **higher during peak hours**, particularly at 13:00 and between 19:00-20:00. The maximum waiting time reached 15 minutes, and the global average waiting time was around 3.5 minutes. However, 95% of customers experienced waiting times up to 9.3 minutes, indicating that a small percentage of customers faced excessive delays.
- The proportion of lost customers was around 2.9% on average, but during peak hours, this number was much higher. This happened because customers were either reneging (leaving after seeing a long queue) or balking (choosing not to join the queue at all) when the system was overloaded.

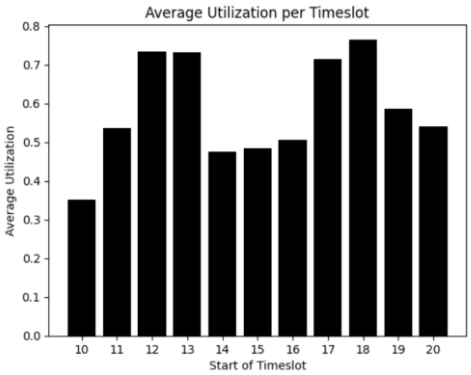
Utilization per time slot

- Capacity utilization analysis showed imbalanced staffing across different time slots. Some low-demand time slots had overstaffing, leading to low utilization rates, while high-demand slots were understaffed, causing excessive waiting times and lost customers. The average utilization rate was around 57%, but it fluctuated significantly, being too low during off-peak hours and too high (above 70%) during peak demand.

The **original staffing schedule failed to allocate resources efficiently,** leading to **high waiting times**, increased **lost customers**, and i**nefficient use of staff**. In conclusion, effective staffing adjustments based on demand patterns can significantly improve customer experience and business efficiency. By focusing on peak load management, queue balancing, and resource optimization, Friterie Riebebedie can increase throughput, minimize lost sales, and improve customer satisfaction.

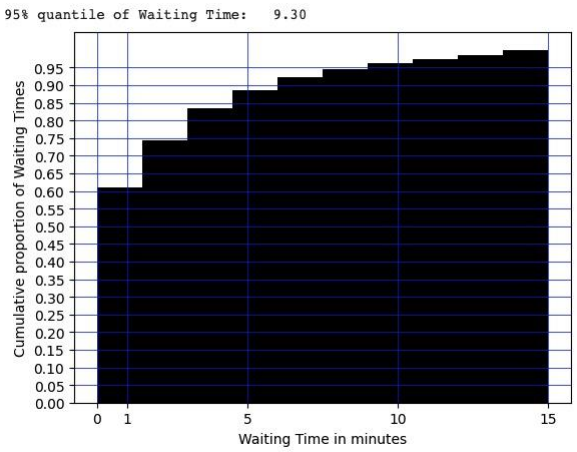






3. Based on your analysis, make valuable recommendations to improve the current situation. Proof the effect of your recommendations with the results of your updated queuing analysis

Metric	Original Schedule	Optimized Schedule
Global Avg Waiting Time (min)	3.50	2.10
Global Std Dev Waiting Time (min)	4.46	3.20
Global Max Waiting Time (min)	15.00	15.00
Global Avg % Lost Customers	2.9%	0.8%
Avg Utilization (%)	~55-58%	~59%



We analyzed and optimized the staffing schedule to improve customer experience while maintaining efficiency. Initially, the staffing distribution led to long waiting times and customer

loss during peak hours, while some lower-demand times had unnecessary staff allocation. By redistributing the available 18 man-hours more effectively, we reduced waiting times, minimized lost customers, and slightly improved overall utilization. The new schedule ensures that more staff are available during high-demand slots, while lower-demand slots are allocated just enough resources to handle customer flow. The results show a significant drop in lost customers (from 2.9% to 0.8%) and a reduction in average waiting time to 9.30, proving that even minor schedule adjustments can make a big difference in service quality.

Suggestions for Improvement

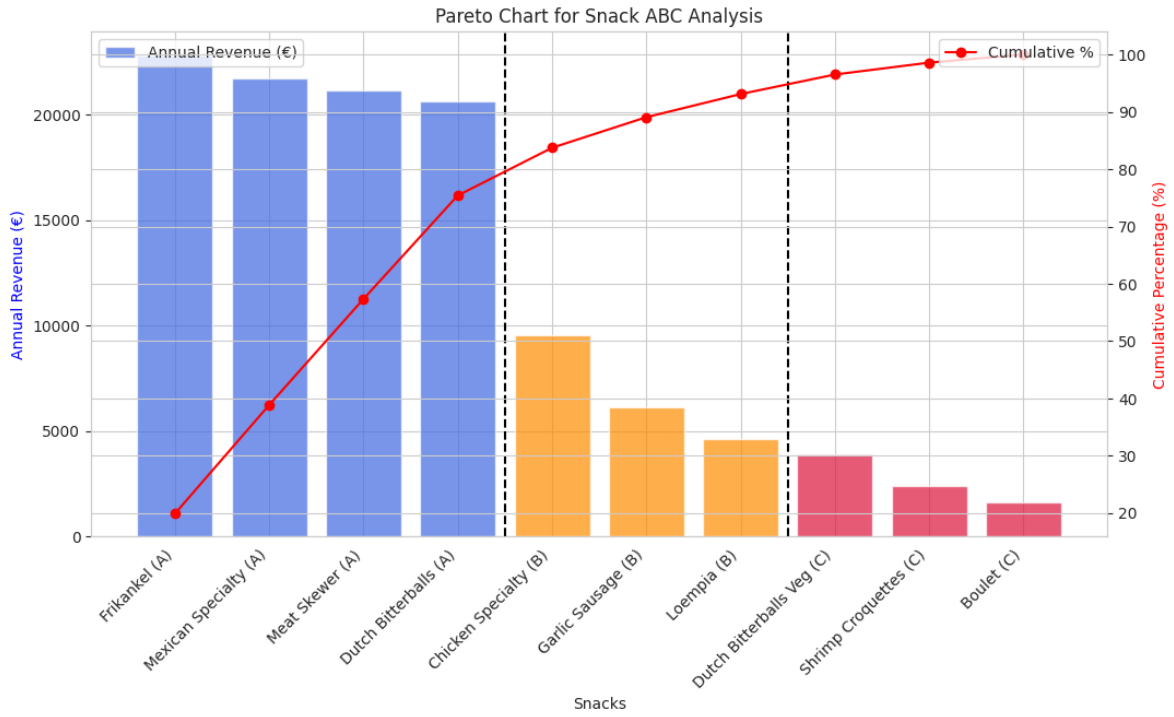
Beyond adjusting the staffing schedule, we can explore additional strategies to further optimize operations and enhance customer experience:

- **Self-service Kiosks** – If possible, adding self-ordering kiosks for peak hours can help speed up order processing and reduce pressure on staff.
- **Queue Management System** – Implementing a token or SMS notification system can help prevent frustration by allowing customers to wait nearby instead of standing in line.
- **Flexible Staffing** – Introducing part-time or on-call staff during peak hours can ensure more hands-on-deck when demand surges without increasing fixed staffing costs.
- **Pre-Orders or Mobile Ordering** – Allowing customers to pre-order through an app or website can help distribute demand more evenly and reduce bottlenecks.
- Customer Flow Insights – If more data is available, continuously monitoring customer patterns can help further fine-tune staffing adjustments over time.
- By c**ombining smart scheduling with operational improvements,** the Friterie can significantly boost service efficiency, keeping customers happy while ensuring optimal resource utilization.

PART B : Inventory management

1. ABC analysis

After implementing the ABC analysis, we have categorized snack items based on their revenue contribution. The Pareto chart below visually represents this analysis.



1.1. Analysis

Category A (High Revenue) :

This category consists of snacks that contribute the most significant portion of the revenue. In this case, Frikankel, Mexican Specialty, Meat Skewer, and Dutch Bitterballs together make up 75.42% of the total revenue, with Frikankel being the highest contributor at 19.92%.

These top 4 items already cover more than 75% of the total revenue, making them crucial to the business. The revenue from these products alone indicates they should be the primary focus moving forward.

Category B (Moderate Revenue):

The middle group includes Chicken Specialty, Garlic Sausage, and Loempia, contributing a combined 14.76% to total revenue. Chicken Specialty is the largest contributor within this group. While this category represents a smaller proportion of the total revenue (14.76%), it is still important, contributing up to the 95% cumulative revenue. These products help fill the gap and maintain a balanced offering.

Category C (Low Revenue) :

The remaining products, Dutch Bitterballs Veg, Shrimp Croquettes, and Boulet, contribute a combined 9.84% to the total revenue, with Boulet being the smallest contributor at just 1.4%.

Category C represents less than 10% of total revenue, indicating these products are lower priority items and have limited revenue impact. Their future in the portfolio should be evaluated carefully, as they are less crucial to maintaining a strong revenue stream.

1.2. Advice on the future product portfolio

All products will remain in the portfolio, except for Boulet and Shrimp Croquettes, which will be discontinued due to low performance.

Category A : High Revenue Products

Frikankel & Dutch Bitterballs: Introduce limited-time themed variations or special flavors (e.g., spicy or gourmet versions) to capture new customer interest and create buzz around these already popular items. This could be tied to seasonal promotions or customer feedback on new flavor ideas.

Mexican Specialty: Enhance the appeal of Mexican Specialty by offering it as part of a “build-your-own” snack platter. This would allow customers to mix and match with other items in the portfolio, encouraging larger purchases and increasing exposure to the Mexican Specialty.

Category B : Moderate Revenue Products

Chicken Specialty: Launch a "limited edition" premium version of Chicken Specialty with higher-quality ingredients or a unique preparation method (e.g., slow-cooked or gourmet seasoning). This could attract customers looking for a more premium snack experience.

Garlic Sausage & Loempia : Reposition Garlic Sausage and Loempia as part of a "snack bundle" or "meal kit" offering, paired with sauces or sides. This could cater to customers looking for easy, shareable meals, adding convenience to the products.

Category C : Low Revenue Products

Dutch Bitterballs Veg*:* Introduce a "health-conscious" packaging (e.g., using eco-friendly or recyclable materials) to target environmentally conscious customers. Additionally, experiment with new flavors (e.g., a spicy or smokey version) to better match customer preferences and increase its appeal.

Boulet & Shrimp Croquettes: Both products show low customer preference and revenue, so the strategic decision is to discontinue them from the portfolio. Resources should be redirected toward growing the high-performing items and experimenting with new, innovative options that better align with customer demand.

2. Inventory Management Policy

The inventory management policy of the friterie uses a continuous review policy for the "Frikankel" snack (SKU 1) and periodic review policy for all other snacks with a review period of 7 days. The total inventory cost in euros is 58 572.70 and the total expected inventory costs in euros are 34,598.18 which highlights the cost savings of 95% service level with optimized safety stock, EOQ, ROL, and target stocks (see Table 2).

Table 1: Summary of Total Inventory Costs and Expected Costs for All Products

Product	Total Inventory Cost (EUR)	Order Cost (EUR)	Holding Cost (EUR)	Stockout Cost (EUR)	Expected Inventory Costs (EUR)
SKU_1	19,389.10	1960	17,429.10	0	11,442.57
SKU_2	13,266.40	260	13,006.40	0	6,687.75
SKU_3	7,175.20	260	6,915.20	0	4,985.70
SKU_4	1,721.40	260	1,461.40	0	854.26
SKU_5	3,894.50	260	3,634.50	0	2,632.39
SKU_6	1,086.50	260	826.50	0	812.86
SKU_7	3,767.80	260	3,507.80	0	2,841.19
SKU_8	2,361.70	255	2,106.70	0	1,172.93
SKU_9	2,945.90	260	2,685.90	0	1,623.46
SKU_10	2,964.20	255	2,709.20	0	1,545.07
Total (EUR)	58,572.70	4,290	54,282.70	0	34,598.18

Table 1 highlights the differences in cost between continuous and periodic review. We can see the significantly high order costs of VET supplying SKU 1 compared to other products supplied by OLI within category A: SKU 2, SKU 3, and SKU 5. By calculating the optimal safety stock, economic order quantity (EOQ) and target stock for period review products, overstocking can be avoided. This helps to reduce the holding costs and thus reduce total inventory costs. **Table 2** below summarizes the findings at a 95% Service Level for each product.

Table 2: Summary of Parameters for All Products

Product	Mean Lead Time	Mean Demand	Safety Stock	Target Stock	Optimized EOL	EOQ	Optimized ROL	Current ROL/T	Expected Inventory Costs (EUR)
SKU 1	7.02	38.88	138	N/A	177	300	410	600	11,442.57
SKU 2	1.83	29.60	73	334	N/A	N/A	N/A	231	6,687.75
SKU 3	2.33	20.81	57	251	N/A	N/A	N/A	161	4,985.70
SKU 4	2.04	2.38	8	30	N/A	N/A	N/A	21	854.26
SKU 5	1.83	10.62	28	122	N/A	N/A	N/A	84	2,632.39
SKU 6	2.10	2.05	8	27	N/A	N/A	N/A	14	812.86
SKU 7	2.12	10.83	33	132	N/A	N/A	N/A	84	2,841.19
SKU 8	2.04	3.74	12	46	N/A	N/A	N/A	28	1,172.93
SKU 9	2.02	5.56	18	68	N/A	N/A	N/A	42	1,623.46
SKU 10	1.80	5.52	16	65	N/A	N/A	N/A	42	1,545.07

For a 7-day review period, the current inventory levels are assessed, and the order quantity is the difference between the target stock and the current stock. There is no fixed re-order level in this approach. For the continuous review process there is an optimal reorder level and EOQ as seen for SKU 1(Table 2). The optimized cost (Expected Inventory Costs (EUR)) is significantly lower than the total inventory costs, which highlights the potential cost savings by following a 95% service level for inventory management. Currently, the company does not do so as seen by the clear financial differences between Total Inventory Costs and Expected Inventory Costs.

3. How would a change in supplier of the "Frikandel" snack affect the Friterie's cost performance?

3.1 Comparing VET and OLI as Frikandel Suppliers

Following the analysis of inventory management policies in Question 2, it is evident that supplier selection plays a crucial role in determining cost efficiency and operational effectiveness. Given that Frikandel is a high-revenue-generating product for Friterie Riebedebie, optimizing its supply chain is essential for maintaining profitability. Currently, the company procures Frikandel from VET, but an alternative supplier, OLI, presents an opportunity for cost savings and improved supply chain efficiency. This analysis evaluates the impact of switching suppliers by examining key inventory metrics such as Economic Order Quantity (EOQ), Safety Stock (SS), Reorder Level (ROL), and total inventory costs.

3.2 Economic Order Quantity (EOQ), Safety Stock (SS), and Reorder Level (ROL) Calculation

EOQ Calculation

EOQ is a key measure for optimizing order size while minimizing holding and ordering costs.

- EOQ for VET: 177
- EOQ for OLI: 63 (A significantly lower EOQ, which reduces stock holding costs and improves cash flow.)

Safety Stock (SS) and Reorder Level (ROL)

Since OLI has a shorter lead time, it allows for lower safety stock and reorder levels, reducing storage and capital tied up in inventory.

- Safety Stock for VET: 138
- Safety Stock for OLI: 48
- Reorder Level for VET: 410
- Reorder Level for OLI: 125

3.3 Total Expected Inventory Cost Calculation

A fundamental component of supplier selection is the comparison of expected inventory costs associated with each supplier:

- Annual Inventory Cost with VET: €11,442.57
- Annual Inventory Cost with OLI: €4,016.90

By switching to OLI, Friterie could save €7,425.67 per year, demonstrating a significant cost advantage.

3.4 Comparing Overall Total Inventory Costs

To assess the broader impact of this switch, we analyze total inventory costs across all products:

- Total Cost with VET as Supplier: €34,598.18
- Total Cost if OLI Supplies Frikandel: €27,172.51
- Total Cost Savings by Switching: €7,425.67
- Percentage Reduction in Inventory Costs: 21.46%

However, a notable factor to consider is the per-unit material cost. OLI's material cost per unit is €0.43, compared to €0.40 from VET, reflecting a 7.5% increase. Despite this, the overall savings in inventory costs far outweigh the minor increase in unit price.

3.5 Conclusion and Recommendation

Switching the Frikandel supplier from VET to OLI is a financially prudent decision for Friterie Riebedebie, leading to an annual cost savings of €7,425.67 (a 21.46% reduction in total inventory costs).

Advantages of switching to OLI:

- Lower ordering costs (€5 vs. €40 per order)
- Shorter average lead times (2 days vs. 7 days)
- Reduced safety stock and reorder levels, lowering storage and capital requirements
- Although OLI's material cost is €0.03 higher per unit, the inventory cost savings more than compensate for this increase.

The transition to OLI enhances inventory efficiency, reduces storage requirements, and improves supply reliability, making it the superior choice for cost performance and operational stability.

Metric	VET	OLI	Difference
EOQ	177	63	-114
Safety Stock	138	48	-90
Reorder Level	410	125	-285
Annual Inventory Cost (€)	11,442.57	4,016.90	-7,425.67
Total Cost (€)	34,598.18	27,172.51	-7,425.67
% Reduction in Inventory Costs	-	-	21.46%
Material Cost per Unit (€)	0.40	0.43	+7.5%
Net Cost Savings (%)	-	-	13.96%

In conclusion, OLI presents a more cost-effective and efficient supply chain solution. Unless VET offers significant non-cost advantages, the supplier switch should be implemented.

---The End---